



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1 – III Semester (2025 – 2026)
Faculty Name	MD Razia
Student Name	NV Karthikeya
Roll Number	2520030307
Branch	CSE
Course Outcome	Co-2

Case Study topic: AirLink Passenger and Flight Record Indexing using B+ Tree

Work Done Summary:

In this case study, we implemented a Flipkart Product Search and Indexing System using a B+ Tree. The B+ Tree stores product IDs in sorted order and supports efficient search, insertion, and retrieval operations. Since all product records are maintained at the leaf level, the system enables fast database indexing and range-based product searches. B+ Trees are widely used in e-commerce platforms to manage large product catalogs efficiently.

Implementation Details:

1. Implemented a B+ Tree to store product IDs.
2. Inserted product records into the tree.
3. Maintained sorted indexing of products.
4. Performed node splitting whenever a node became full.
5. Stored all product records in leaf nodes.
6. Implemented efficient search operations.
7. Supported range queries through linked leaf nodes

Results and Screenshots:



The Flipkart Product Search System was successfully implemented using a B+ Tree. Product records were indexed efficiently, enabling fast search and retrieval operations. The search for Product ID 1020 was completed successfully, demonstrating efficient database indexing.

GitHub Link: <https://github.com/viswanadhakarathikeya-dot/DSA-2>

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Case Study topic: AirLink Airport Connectivity Optimization using Minimum Spanning Tree (MST)

Work Done Summary:

In this case study, we implemented an Electric Power Grid Optimization System using the Minimum Spanning Tree (MST) concept. Power stations are represented as graph vertices, and transmission lines are represented as weighted edges. Kruskal's Algorithm was used to connect all power stations with the minimum installation cost while avoiding cycles. The MST ensures an optimized and cost-effective power distribution network.

Implementation Details:

1. Represented power stations as graph vertices.
2. Represented transmission lines as weighted edges.
3. Applied Kruskal's Algorithm to construct the MST.
4. Sorted edges based on transmission cost.
5. Used Union-Find for cycle detection.
6. Connected all power stations with minimum cost.
7. Calculated the total optimized network cost.

Results and Screenshots:

```
Shortest Flight Path from Srinagar (SXR) to Kanyakumari (TRV)
Step-by-Step Distance Calculation

[0] SXR (Srinagar) - 0 km
|
| Step 1: Fly to DEL
| 800 km
v
[0] DEL (Delhi) - 800 km
|
| Step 2: Fly to NAG
| 800 + 1000 = 1800 km
v
[0] NAG (Nagpur) - 1800 km
|
| Step 3: Fly to HYD
| 1800 + 500 = 2300 km
v
[0] HYD (Hyderabad) - 2300 km
|
| Step 4: Fly to BLR
| 2300 + 600 = 2900 km
v
[0] BLR (Bengaluru) - 2900 km
|
| Step 5: Fly to TRV
| 2900 + 700 = 3600 km
v
[0] TRV (Kanyakumari/Trivandrum) - 3600 km

*Alternative Route via MAA (Chennai): 3630 km (Longer)*
Final Shortest Path: SXR -> DEL -> NAG -> HYD -> BLR -> TRV = 3600 km
```

SCENARIO / CASE STUDY

Airlink Flight-Route shortest path with air-traffic penalties.

Airlink operates a route network with 7 airport hubs {DEL (Delhi), BOM (Mumbai), HYD (Hyderabad), CCU (Calcutta), BLR (Bangalore), MAA (Madurai), AMD (Amritsar)}. Edge weights are average flight times (minutes) plus an air traffic congestion penalty factor.

The dispatcher needs to compute the shortest travel time from DEL to every other hub for this network. This means Dijkstra's algorithm cannot be used naively. Bellman-Ford handles negative edges.

Directed Airlink route graph · 11 edges · weights = flight time + congestion penalty (air traffic congestion). One edge (BLR->MAA) has weight -3 (Tailwind bonus).

Source = DEL. Negative edge (red dashed) means Dijkstra is not safe; use Bellman-Ford.

Dijkstra's Shortest Path Algorithm (AirLink Example)

Step 1: Initialization

Start at DEL (Delhi) with distance 0. Others set to ∞ (infinity).

Step 2: Relax & Update Neighbors

Relax DEL's neighbors:

- BOM = 10
- AMD = 40

Step 3: Pick Next Node

Select BOM (10).

- Update HYD = 25
- Update CCU = 30

Final Result (Treat BLR->MAA as 0)

MAA Distance: 55 (Incorrect).

▲ Should be 52 due to the negative tailwind edge!

Dijkstra FAILS on negative edges!

The Electric Power Grid Optimization System was successfully implemented using Kruskal's Minimum Spanning Tree algorithm. The MST connected all power stations with the minimum transmission cost while avoiding cycles. The optimized network achieved a minimum total cost of 9 units.

GitHub Link: <https://github.com/viswanadhakarathikeya-dot/DSA-2>

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